CHE 2108 Physical Chemistry Laboratory.
Revision Quiz 01.

1. What is Beer's Lambert law, give formula and define the terms.
2. What are the limitations of Beer's Lambert law.
3. Suggest a practical to determine unknown Fe^{2+} content in waste water sample using following materials and instruments.
   ($KMnO_4$, Calomel electrode , Pt electrode, Digital multi meter , Connecting wires and you may use any glass wares)
4. Define the terms of Arrhenius equation.
5. Rearrange the above equation, reaction rate constant vs temperature.
6. Give the factors for affecting rate of the reaction.
7. Give the expression for pH of Quinhydron electrode.
8. The absorbance of an iron thiocyanate solution containing 0.00500 mg Fe/mL was reported as 0.4900 at 540 nm.
   a. Calculate the specific absorptivity, including units, of iron thyocyanate on the assumption that a 1.00 cm cuvette was used.
   b. What will be the absorbance if (1) the solution is diluted to twice its original volume and (2) the solution is placed in a 5.00 cm cuvette?

9. The concentration of dye compound in an aqueous solution is 10 µg/mL. The absorbance is found to be 0.209 when this solution is placed in a 1.00 cm cuvette and 258 nm radiation is passed through it.

   a. Calculate the specific absorptivity, including units, of dye compound
   b. What will be the absorbance if the solution is diluted to 1µg/mL?
   c. What will be the absorbance if the path length of the original solution is increased to 5.00 cm?

10. A 2.845 g sample of an unknown compound containing cobalt(II) nitrate (MW=182.943 g/mol) was dissolved in water in a 250 mL volumetric flask. The absorbance of this solution was found to be 0.2195 at 515 nm in a 1 cm cell. The molar absorptivity of the given compound is 4.8 $moldm^3cm^{-1}$. Calculate the percentage of cobalt nitrate in the unknown sample.